Novel Land-Cover Classification Approach With Nonparametric Sample Augmentation for Hyperspectral Remote-Sensing Images

Zhiyong Lv, *Senior Member, IEEE*, Pengfei Zhang, Weiwei Sun, *Senior Member, IEEE*, Jón Atli Benediktsson, *Fellow, IEEE*, and Tao Lei, *Senior Member, IEEE*

***Abstract*— Samples play a crucial role in the supervised classification of remote-sensing images. However, labeling large samples for training a classifier or deep-learning network is not only time-consuming, but also labor-intensive. In this article, a novel land-cover classification with nonparametric sample augmentation is proposed to improve the performance of hyperspectral remote-sensing image (HRSI) classification. First, initial samples with limited quantity are selected randomly from the ground-truth map. Second, based on the gray image, a nonparametric adaptive region generation (NARG) algorithm is developed for utilizing the contextual information around each sample. Then, a nonparametric sample augmentation algorithm is developed with NARG to explore reliable samples iteratively around each initial sample. Finally, the above steps are fused into an iterative process to obtain the final classification map. Compared with some typical traditional methods and some widely used deep-learning methods based on four real HRSIs, our proposed approach exhibits some advantages in improving the visual performance and quantitative accuracies of HRSI classification, such as the improvement is about 2.0%–10.34% for four real HRSIs in terms of the overall accuracy (OA).**

***Index Terms*— Land-cover classification, limited samples, remote-sensing image, sample augmentation.**

I. Introduction

LAND-COVER classification with remote-sensing images aims to assign a label for each pixel [1], [2], [3]. Owing to the advantages of acquiring remote-sensing images with drones or satellites, such as observing the land-cover information in a large area with a lower time cost, land-cover classification with remote-sensing images plays an important role in land-use analysis [4], cropland mapping [5], and environmental health evaluation [6], [7].

In recent decades, with the quick development of remote-sensing technology, various images, including hyperspectral, low-median spatial resolution, and high-resolution images, have become conveniently available. However, the abundance of remote-sensing image sources does not mean high classification accuracies and more applications in practical engineering because the Hughes phenomenon and mixed pixels are popular for remote-sensing images [8], [9]. In particular, due to the high-dimensional characteristics and mixing pixels of hyperspectral remote-sensing images (HRSIs), the task of labeling samples becomes time-consuming and challenging.

Many researchers have concentrated on developing classification methods with small training samples. Here, these methods are briefly divided into traditional and deep-learning-based methods in the literature review, and their details are presented below.

The traditional method of classification with small samples is defined by generating new samples using an initial small sample set to improve the classification performance. A typical way of applying traditional methods is using the synthetic minority oversampling technique (SMOTE) to generate new samples [10]. SMOTE concentrates on oversampling the minority class to deal with the problem of imbalanced classification, and it has outstanding advantages in terms of simplicity and robustness [11], [12], [13], [14]. However, SMOTE generates new samples linearly that cannot describe the distribution of the real sample [15]. In addition, utilizing the neighboring context around each initial sample is also beneficial for exploring potential similar samples. For example, Richards and Jia [16] augmented training samples by using suitable neighbors to enhance HRSI classification performance, Lv et al. [17], [18] explored potential samples around each initial sample iteratively to enhance the very high-resolution remote-sensing image classification performance, and Lv et al. [19] proposed iterative training sample augmentation method to explore the potential samples around each original sample. Moreover, some researchers have noted a negative effect on classification performance from mislabeled samples. For example, Foody and Mathur [20] and

Manuscript received 20 March 2023; revised 30 April 2023 and 27 June 2023; accepted 27 August 2023. Date of publication 30 August 2023; date of current version 11 September 2023. This work was supported in part by the Open Fund of Key Laboratory of Urban Land Resources Monitoring and Simulation, Ministry of Natural Resources, under Grant KF-2022-07-010; in part by the National Natural Science Foundation of China under Grant 41971296, Grant 42122009, and Grant 42271385; in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR19D010001; in part by the Public Science and Technology Plan Projects of Ningbo City under Grant 2021S089; in part by the Science and Technology Innovation 2025 Major Project of Ningbo City under Grant 2021Z107 and Grant 2022Z032; and in part by the "Pioneer" and "Leading Goose" Research and Development Projects of Zhejiang Province under Grant 2023C01027. *(Corresponding authors: Weiwei Sun; Zhiyong Lv.)*

Zhiyong Lv and Pengfei Zhang are with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: Lvzhiyong_fly@hotmail.com; Zhangpengfei_life@hotmail.com).

Weiwei Sun is with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: nbsww@outlook.com).

Jón Atli Benediktsson is with the Faculty of Electrical and Computer Engineering, University of Iceland, 107 Reykjavik, Iceland (e-mail: benedikt@hi.is).

Tao Lei is with the Shaanxi Joint Laboratory of Artificial Intelligence, School of Electronic Information and Artificial Intelligence, Shaanxi University of Science and Technology, Xi'an 710021, China (e-mail: leitao@sust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3309949

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Kang et al. [21] proposed mislabeled sample detection and correction methods to improve HRSI classification accuracies. Many traditional methods have demonstrated that classification performance with HRSIs can be improved by enhancing the quality and quantity of the sample. However, traditional methods of sample augmentation usually require empirical parameters and the optimal setting needs trial-and-error experiments, which are time-consuming.

Besides traditional methods for classification with small samples, deep-learning techniques have become a hot topic in the field of HRSI classification [22]. First, the most widely used convolutional neural networks (CNNs), including CNN-based contextual information extraction [23], 2-D CNN [24], and 3-D learning CNN framework [25], are popular in HRSI classification. However, although these deep-learning methods can achieve satisfactory classification performance with remote-sensing images, owing to the fact that different targets have various shapes and sizes, deep learning with a fixed convolutional kernel cannot cover the various targets precisely.

Therefore, to further improve classification accuracy, many researchers have designed multiscale neural networks to cover various ground targets with different sizes and shapes, such as the deep pyramidal residual network (DPY-ResNet) [26], multiscale residual network [27], adaptive spectral–spatial feature extraction [28], [29], and spectral–spatial multiscale network [30]. Moreover, adaptive context plays an important role in hyperspectral image classification, such as [adaptive spectral-spatial multiscale network (ASSM Net)] [31], adaptive multiscale deep fusion for image classification [32], adaptive spectral-spatial kernel ResNet (A2S2K-Net) [33], and the classical deep residual learning net (Res-Net) [34]. From the experimental results obtained by the state-of-the-art deep-leaning methods, it can be found that deep-learning methods can obtain a satisfying classification map. However, these methods usually require a large number of samples to train a neural network.

In recent years, lots of researchers concentrated on deep-learning methods with small training samples. For example, the deep few-shot learning network (DFSL-Net) [35], [36], multitask deep learning (MDL) [37], [38], and limited training sample-based learning [39], [40], [41] have promoted for land-cover classification performance with HRSIs using few samples. In these approaches, the importance of the quantity and quality of samples has been noted in improving land-cover classification performance with HRSIs [42]. Moreover, various mechanisms, such as the spectral–spatial Siamese network (S3Net) [36] and MDL [37], are recommended for enhancing classification performance while the number of training samples is limited. One advantage of the approaches lies in requiring few samples yet achieving satisfied classification performance.

In this article, we aim to improve the classification performance with HRSIs through nonparametric sample augmentation. The motivation of the proposed approach has three aspects: generating an adaptive spatial domain around each initial sample by measuring spectral similarity, exploring the reliable potential samples in the adaptive spatial neighboring domain with refining rules, and balancing the user's accuracy by iteratively adjusting the quantity of the sample for each class. The experimental results based on four real HRSIs demonstrate the advantages of our proposed approach in improving classification performance and accuracy. The major contributions of the proposed approach are summarized as follows.

1) A nonparametric adaptive region generation (NARG) algorithm is promoted to utilize contextual information in HRSIs. Intuitively, the shape and size of different individual land-cover patches are unknown in an image scene, and the advantages of utilizing spatial-contexture information adaptively have been verified for land-cover classification [43], change detection [44], [45], [46], and building extraction [47]. However, determining optimal parameter settings for an adaptive domain usually requires trial-and-error experiments for a specific dataset or depends on empirical values. In this study, the NARG algorithm is developed by utilizing the relationship between an initial sample and its neighbors in terms of spectral similarity. Compared with determining the contextual domain with an empirical parameter setting, the promoted NARG algorithm may be more preferable for practical applications.
2) A nonparametric sample augmentation algorithm is developed for mining potential samples around each initial sample. Generally, many existing approaches created new samples that require empirical parameters, and the reliability of the generated new samples for enhancing the classification performance is considered insufficient. Therefore, developing a nonparametric sample-explored algorithm based on a data-driven mechanism is significant for sample augmentation in practical applications.
3) A novel classification approach with nonparametric sample augmentation is proposed for improving the classification performance of HRSIs and balancing the user's accuracy of each class. In general, deep-learning modules were trained with equal patches in each iterative epoch; this may result in cutting down the classified ability of the trained classifier or module. Here, we designed an iterative classification approach to improve the classification performance and balance the user's accuracy. In the proposed approach, limited initial samples are augmented and adjusted iteratively, and then the renewed samples are utilized correspondingly for classification.

The rest of this article is organized as follows. The proposed approach is detailed in Section II. The experimental results are presented in Section III, and the conclusion is given in Section IV.

II. Methodology

A. Overview of the Proposed Approach

In this section, we first present the overview of the proposed classification approach with nonparametric sample augmentation. Fig. 1 shows the flowchart for the proposed approach, it consists of three major parts: 1) a NARG algorithm; 2) a nonparametric sample augmentation approach; and

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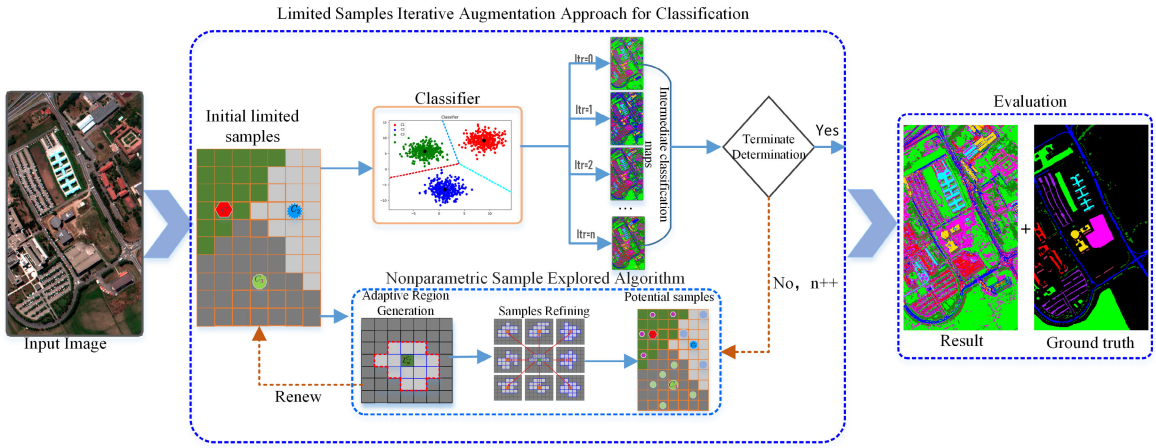


Fig. 1. Flowchart of the classification approach with nonparametric sample augmentation for HRSIs.

3) termination condition for the iteration of the proposed classification approach. Details are presented as follows.

B. Nonparametric Adaptive Region Generation Algorithm

Many studies with remote-sensing images have indicated that using spatial context adaptively has advantages in smoothing salt-and-pepper noise and preserving the edges of ground targets with various shapes and sizes [48], [49], [50]. However, optimal parameter settings to determine an adaptive region around each pixel in a given image are usually determined through trial-and-error experiments, which are time-consuming [51], [52]. In this section, a NARG algorithm is proposed to determine an adaptive region around each initial sample. Algorithm 1 shows the details of the proposed NARG.

Based on the details of Algorithm 1, two aspects are worth noting: 1) an adaptive region around an initial sample can be generated without any predefined parameters and 2) the extended adaptive region is composed of a group of pixels characterized by spectral similarity and spatial continuity. Therefore, utilizing spatial contextual information around an initial sample in an adaptive manner may be more precise than using a regular window or mathematical model for exploring potential samples.

C. Nonparametric Sample Augmentation Approach

Using the proposed NARG algorithm can generate an adaptive region around each initial sample for exploring potential samples. In this section, a nonparametric sample augmentation approach is developed based on the NARG for amplifying initial small samples, and its details are presented as follows.

1) To be clarified clearly, some symbols are first defined: an initial sample and its adaptive region are denoted as $S_{i,j}^{c_n^l}$ and $R_{i,j}$, respectively. The pixels within $R_{i,j}$ are given as $P_{m,n}$, where $m \in R_{i,j}$ and $n \in R_{i,j}$, and the adaptive region around each $P_{m,n}$ is marked as $R_{m,n}$, where $m \neq i$ and $n \neq j$. The standard deviation for $R_{m,n}$ and $R_{i,j}$ is calculated and recorded as $\mathrm{Std}_k^{R_{m,n}}$ and $\mathrm{Std}^{R_{i,j}}$.
2) If $||\mathrm{Std}^{R_{i,j}} - \mathrm{Std}_k^{R_{m,n}}|| < \varepsilon_1$ is true, then the corresponding seed pixel $P_{m,n}$, which is centered at $R_{m,n}$, is selected as a potential sample.

Algorithm 1 NARG

Input: An image (I_1) and the initial samples $S_{i,j}^{c_n^l}$ for each class, where $S_{i,j}^{c_n^l} \in I_1$, c_n^l denotes nth sample for class l.
Output: An adaptive region $R_{i,j}$ around the sample $S_{i,j}^{c_n^l}$.
Step-1: The mean and standard deviation of each initial sample set is calculated for class l, symbolized as $M^{c_N^l}$ and $Std^{c_N^l}$, where N is the total number of samples for the class.
Step-2: $S_{i,j}^{c_n^l}$ is placed into a stack STK, labeled $S_{i,j}^{c_n^l}$ as one pixel of $R_{i,j}$.
Step-3:
While $||M_{i+1}^{c_N^l} - M_i^{c_N^l}|| < \varepsilon_0$ AND $||Std_{i+1}^{c_N^l} - Std_i^{c_N^l}|| < \varepsilon_0$:
1. Pop one element $(E_{i,j})$ from STK.
2. $E_{i,j}$ is compared with P_{sur} (where $P_{sur} \notin R_{i,j}$), which is one pixel of the 8-connected neighbors around $E_{i,j}$:
3. IF $P_{sur} \in [M^{c_N^l} - 2 \times Std^{c_N^l}, M^{c_N^l} + 2 \times Std^{c_N^l}]$ in terms of the gray value is TRUE, then:
3.1. label P_{sur} as one pixel of $R_{i,j}$;
3.2. push P_{sur} into STK;
3.3. update the value of $M^{c_N^l}$ and $Std^{c_N^l}$ with P_{sur}, and record as: $M_{i+1}^{c_N^l}$ and $Std_{i+1}^{c_N^l}$.
ELSE: Go to Step-4.
End while
Step-4: Return $R_{i,j}$.

3) When all the pixels within $R_{i,j}$ are compared and refined as shown in Step-2, a potential sample set is obtained and symbolized as $\mathrm{PS} = \{\mathrm{ps}_1, \mathrm{ps}_2, \mathrm{ps}_3, \ldots, \mathrm{ps}_q\}$, where is denoted as the total number of potential samples. However, the potential sample set may contain different classes owing to the spectral noise of the intraclass in remote-sensing images.
4) To further improve the reliability of the potential samples and cover the spectral variation within a class, some potential samples are selected by an empirical rule in statistics [53]. The empirical rule states that for normal distributions, 68% of observed data points will lie inside one standard deviation of the mean, 95% will fall within

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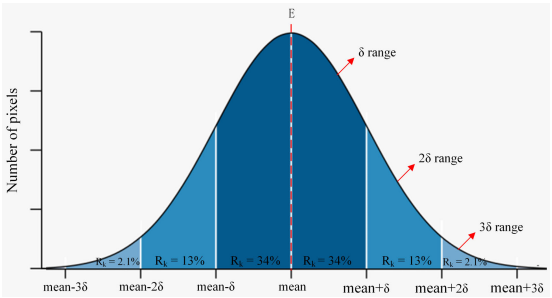


Fig. 2. Demonstration of a pixel distribution and potential samples selection in an adaptive region. The same color indicates the symmetrical range on both sides of the mean.

two standard deviations, and 99.7% will occur within three standard deviations. Here, the distribution of the pixels within an adaptive region around an initial training sample is assumed to be normal. As shown in Fig. 2, the pixel distribution of an adaptive region is divided into three ranges which are named 3δ, 2δ, and δ, and they are marked symmetrically on two sides of the expected value (E). Then, the number of potential samples within each range is calculated by $N_i = \lfloor (n_i/q) \times R_k \rfloor$, where N_i is the number of selected samples in a division, q is the total potential samples in the six ranges, and R_k is a probability between a specific range data and the total dataset for a standard normal distribution, and its value for each division is given in Fig. 2. The motivation for selecting potential samples from different ranges for a specific class lies in covering the spectral homogeneity difference for the intraclass.

From the preceding details of the proposed sample augmentation approach, the advantages of the proposed approach for sample augmentation are briefly summarized as follows: 1) the proposed approach is triggered and driven by initial samples and can achieve sample augmentation without parameter settings; 2) the reliability of the selected samples is guaranteed by refining it twice based on NARG; and 3) the final potential samples around an initial sample is selected from a different range of potential samples, which helps improve the representability of the intraclass samples.

D. Termination Condition for the Iteration of the Proposed Classification Approach

In the proposed iterative classification approach, the initial sample is augmented iteratively and the classification map is obtained correspondingly with the augmented samples. The iteration is terminated until the classification maps from adjacent iterations are similar enough for each class. Details of the terminated condition of the proposed classification approach are defined in the following equation:

$$\left| M_{k-1,k}^{C^l} - M_{k,k+1}^{C^l} \right| < \varepsilon_1 \tag{1}$$

$$M_{k-1,k}^{C^l} = \frac{\text{count}\left(M_{k-1}^{C^l} \leftrightarrow M_k^{C^l} \right)}{W \times H} \tag{2}$$

$$M_{k,k+1}^{C^l} = \frac{\text{count}\left(M_k^{C^l} \leftrightarrow M_{k+1}^{C^l} \right)}{W \times H} \tag{3}$$

where $M_{k-1,k}^{C_l}$ is the ratio between the matched pixels and total pixels in terms of the classification map from the $(k-1)$th and kth iterations for the class C^l. In addition, ε_1 is a very small constant, and it is fixed at 0.003 in our study. Thus, (1) implies that when the difference among the classification maps (M_{k-1}, M_k, and M_{k+1}) from three adjacent iterations ($k-1$, k, and $k+1$) for the specific class C^l is less than ε_1, the sample augmentation for the class C^l is terminated, where W and H are the width and height of an image, respectively, and the total comparative pixels between two adjacent classification maps are $W \times H$ pixels. When the matched pixels of the classification maps for all the classes satisfied the predefined condition (1), the sample iterative augmentation for all the classes is terminated, and then the proposed iterative classification approach is also terminated.

From the perspective of practical application, the major advantages of the proposed classification approach are the following. First, the proposed approach is highly automatic for achieving classification tasks with HRSIs. Although the minimal constants are required for terminating the iterations, they are optional and can be given a default value without hard-tuning. Moreover, competitive accuracies can be achieved under very small initial samples and default parameter settings for different datasets. Second, the user's accuracy of each class can be balanced by adjusting the samples of each class in the iterative process of the proposed classification approach, and the iteration is terminated until all the classes' accuracies meet our predefined condition [see (1)].

III. Experiments

In this section, some experiments are designed to verify the performance of the proposed classification approach based on nonparametric sample augmentation with HRSIs. Details of the experiments are introduced as follows.

A. Dataset Description

Four real HRSIs are used for the experiments. As the visual performance in Fig. 3 and details of each dataset are given in Table I, the four datasets are acquired with the reflective optics system imaging spectrometer (ROSIS) and the NASA airborne visible/infrared imaging spectrometer (AVIRIS) instruments. Moreover, the spatial and spectral resolutions of these datasets are in a wide range from 1.3 to 18.0 m/pixel. The datasets covered different ground scenes, such as urban area (Data-1), city center (Data-2), crops area (Data-3), and beach area (Data-4). Therefore, achieving land-cover classification maps with very limited initial samples for the datasets, which involve different resolutions, sensors, and scenes, is challenging.

B. Experimental Designations

Two experiments are designed to verify the advantages of the proposed approach. These experiments are described as follows.

1) We first compared our proposed methods with achieving classification using the initial sample directly, Richard's method [16], which enhanced the training samples by

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TABLE I
DETAILED DESCRIPTION OF EACH DATASET

Dataset	Location	Size (pixel)	Sensors	Resolution (m/pixel)	Selected bands (R-G-B bands)
Pavia-University (Data-1)	Italy, Pavia	340×610	ROSIS sensor	1.3	103 (60,27,17)
Pavia-Center (Data-2)	Italy, Pavia	715×1096	ROSIS sensor	1.3	102 (60,27,17)
Salinas (Data-3)	America, California	217×512	AVIRIS sensor	3.7	224 (58,37,8)
KSC (Data-4)	America, Florida	614×512	AVIRIS sensor	18.0	224 (52,27,17)

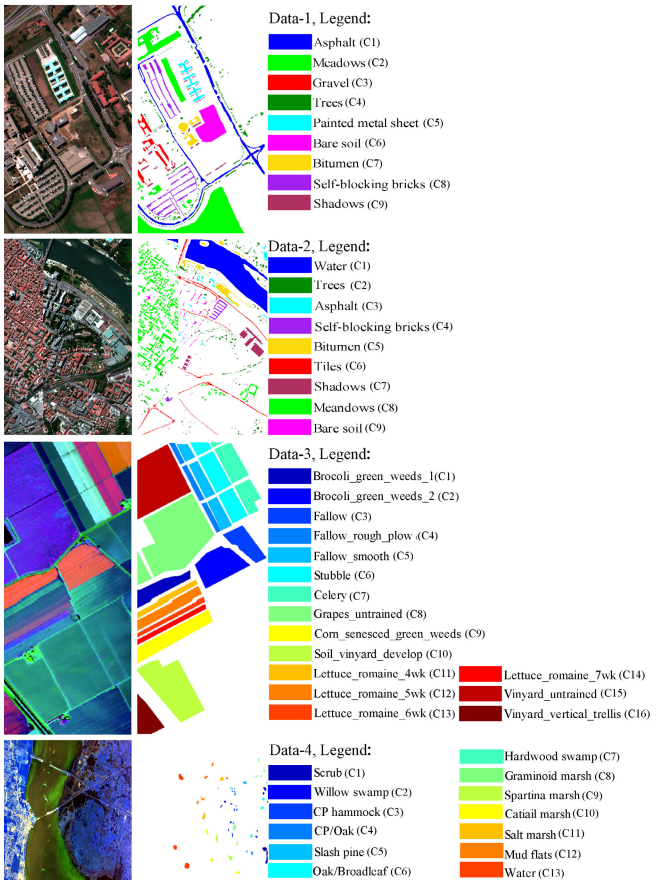


Fig. 3. Experimental datasets visual performance. Data-1–Data-4 are Pavia University HRSI, Pavia Center HRSI, Salinas HRSI, and Kennedy Space Center (KSC) HRSI, respectively. The false color image, ground truth, and legend are arranged from left to right in each row.

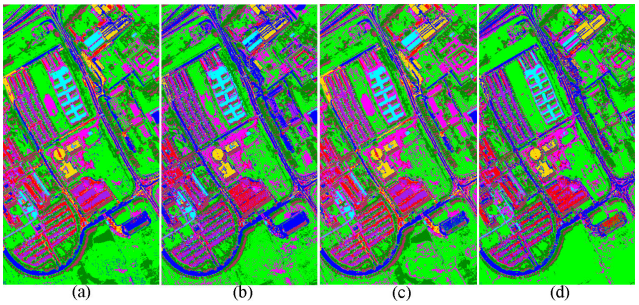


Fig. 4. Acquiring classification maps from different traditional methods for Data-1. (a) Initial samples, (b) Richard's [16], (c) Kang's [21], and (d) our proposed approach.

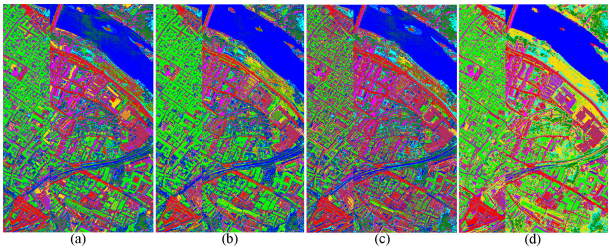


Fig. 5. Acquired classification maps from different traditional methods for Data-2. (a) Initial samples, (b) Richard's [16], (c) Kang's [21], and (d) our proposed approach.

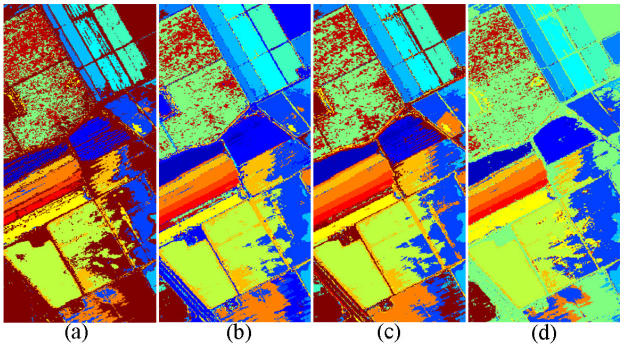


Fig. 6. Acquired classification maps from different traditional methods for Data-3. (a) Initial samples, (b) Richard's [16], (c) Kang's [21], and (d) our proposed approach.

exploring suitable neighbors; and Kang's method [21], which concentrated on detecting and correcting the mislabeled training samples to improve the classification performance with HRSIs. To guarantee the comparative analysis, we set the initial samples for each method as five samples/class for Data-1 and ten samples/class for Data-2, Data-3, and Data-4. Moreover, the parameters of each method are optimized by trial-and-error experiments. For example, the number of initial samples is increased from 1 to 20 pixels per class to determine the least number of initial samples requirement of our proposed approach for each dataset by comparing the corresponding classification map in terms of classification accuracies. In addition, if a method has more than one parameter, one of the parameters will be varied, while other parameters are fixed at an empirical value, and the classification map will be obtained and compared at each change of the parameter's value, and

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TABLE II

COMPARISON BETWEEN SELECTED TRADITIONAL METHODS AND PROPOSED APPROACH FOR EACH DATASET WITH SVM CLASSIFIER (# NUMBER DENOTES THE TOTAL ITERATIONS OF THE PROPOSED APPROACH, Ka $\in [-1, +1]$, OTHERS MEASURED IN PERCENT)

Dataset	Method	OA	AA	Ka	F1-score	SDUA
Data-1	Initial Sample	67.59	64.61	0.575	65.37	21.48
	Richard's Method [16]	70.14	68.75	0.615	71.07	19.64
	Kang's Method [21]	64.97	64.79	0.550	64.96	18.97
	Proposed Method (#13)	**78.34**	**76.83**	**0.719**	**78.55**	**12.02**
Data-2	Initial Sample	79.17	62.13	0.714	64.00	23.43
	Richard's Method [16]	88.47	76.55	0.834	73.36	22.05
	Kang's Method [21]	77.17	66.09	0.691	66.07	22.62
	Proposed Method (#16)	**94.76**	**84.87**	**0.926**	**86.32**	**10.82**
Data-3	Initial Sample	72.17	86.34	0.695	79.37	12.86
	Richard's Method [16]	83.46	87.83	0.815	87.79	13.79
	Kang's Method [21]	76.22	82.11	0.736	80.45	17.70
	Proposed Method (#14)	**86.53**	**90.23**	**0.830**	**90.54**	**10.00**
Data-4	Initial Sample	65.00	56.68	0.612	56.05	23.64
	Richard's Method [16]	76.15	66.83	0.734	65.65	25.07
	Kang's Method [21]	61.93	50.70	0.578	50.27	27.55
	Proposed Method (#13)	**77.20**	**70.57**	**0.746**	**70.48**	**16.06**

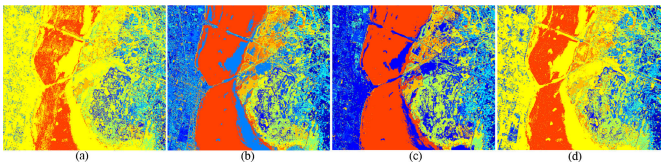


Fig. 7. Acquired classification maps from different traditional methods for Data-4. (a) Initial samples, (b) Richard's [16], (c) Kang's [21], and (d) our proposed approach.

then acquiring the optimal value of the testing parameter. All the methods with the same initial samples are coupled with support vector machines (SVMs) to achieve the performance comparison. The parameters of SVMs used in the following experiments are $C = 0.001$ and Gamma $= 0.001$, and the classifier SVM was trained with the radial basis function (RBF) kernel and fivefold cross-validation.

2) To further demonstrate the superiority of our proposed approach, we compared it with some deep-learning classification methods, including CNN [23], SSRN-Net [25], DPY-ResNets [26], ASSM-Net [31], SSR-Net [39], ResNet [34], A2S2K-Net [33], DFSL-Net [35], MDL4OW [37], and s3-Net [36] on the four datasets. All the parameters of the deep-learning networks are optimized through trial-and-error experiments. In addition, all the applied neural networks are based on the same initial training samples for network training to guarantee the fairness of the comparison. The learning rate and epochs are fixed at 0.001 and 100, respectively, for each deep-learning method to guarantee fairness.
3) In all the experiments, quantitative comparisons are evaluated in terms of three widely used accuracy indicators: overall accuracy (OA), average accuracy (AA), Kappa coefficient (Ka), and F1-score. In addition, to investigate the user's accuracy-balanced ability for each class, the standard deviation of the user's accuracy (SDUA) is calculated for each result. A smaller SDUA means higher consistency of the user's accuracy for all the classes

$$\text{SDUA} = \sqrt{\frac{1}{N}\sum_{1}^{N}(x_i - \mu)^2}, \quad \text{where } \mu = \frac{1}{N}\sum_{1}^{N} x_i \tag{4}$$

where SDUA is the standard deviation of the user's accuracy. As mentioned before, N is the total number of interesting classes in an image scene, x_i is the user accuracy for the ith class, and μ is the mean value of the user's accuracy for all the classes.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE SELECTED DEEP-LEARNING METHODS AND PROPOSED APPROACH WITH THE SVM CLASSIFIER FOR DATA-1 (# NUMBER DENOTES THE TOTAL ITERATIONS OF PROPOSED APPROACH, Ka $\in [-1, +1]$, OTHERS MEASURED IN PERCENT)

Methods	OA	AA	Ka	F1-score	SDUA
CNN [23]	73.36	69.82	0.653	66.25	30.19
SSRN-Net [25]	75.65	70.46	0.691	69.70	16.74
DPY-ResNet [26]	57.84	59.13	0.482	56.52	27.12
ASSM-Net [31]	56.35	54.65	0.462	50.25	31.86
SSR-Net [39]	70.96	66.56	0.631	66.94	17.20
Res-Net [34]	56.38	64.94	0.468	56.08	31.37
A2S2K-Net [33]	76.34	77.64	0.693	75.24	16.13
DFSL-Net [35]	77.84	77.67	0.713	77.75	13.81
MDL-Net [37]	75.16	**78.60**	0.694	78.20	12.46
S3-Net [36]	76.93	76.89	0.708	78.34	14.45
Proposed Method (#13)	**78.34**	76.83	**0.719**	**78.55**	**12.05**

C. Experimental Results

With the above experimental designations, experimental results based on four real HRSIs are obtained. Details are presented below.

1) As shown in Table II, the proposed approach achieved the best accuracies in terms of OA, AA, Ka, F1-score, and SDUA, as shown by comparison with the traditional

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TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE SELECTED DEEP-LEARNING METHODS AND THE PROPOSED APPROACH WITH THE SVM CLASSIFIER FOR DATA-2

Methods	OA	AA	Ka	F1-score	SDUA
CNN [23]	86.74	73.73	0.816	66.25	14.55
SSRN-Net [25]	78.94	80.51	0.699	71.25	21.94
DPY-ResNet [26]	70.98	60.87	0.594	40.89	19.62
ASSM-Net [31]	81.23	76.11	0.733	63.82	17.39
SSR-Net [39]	86.03	64.40	0.799	48.09	15.03
Res-Net [34]	69.49	51.19	0.592	46.34	21.11
A2S2K-Net [33]	89.64	78.65	0.853	72.25	14.56
DFSL-Net [35]	93.09	80.11	0.903	82.70	12.41
MDL-Net [37]	93.53	81.03	0.910	83.61	12.46
S3-Net [36]	94.52	84.31	0.923	86.07	11.50
Proposed Method (#16)	**94.76**	**84.87**	**0.926**	**86.32**	**10.82**

TABLE V

QUANTITATIVE COMPARISON BETWEEN THE SELECTED DEEP-LEARNING METHODS AND THE PROPOSED APPROACH WITH THE SVM CLASSIFIER FOR DATA-3

Methods	OA	AA	Ka	F1-score	SDUA
CNN [23]	82.31	90.65	0.801	85.88	10.60
SSRN-Net [25]	83.71	87.93	0.819	83.64	14.66
DPY-ResNet [26]	83.84	86.29	0.821	87.82	11.98
ASSM-Net [31]	79.90	90.96	0.779	88.98	12.99
SSR-Net [39]	75.36	76.90	0.728	72.91	15.88
Res-Net [34]	79.54	87.58	0.775	84.02	16.26
A2S2K-Net [33]	84.73	89.64	0.830	90.17	11.80
DFSL-Net [35]	84.00	**90.40**	0.823	89.99	13.39
MDL-Net [37]	82.95	88.14	0.800	87.79	13.13
S3-Net [36]	85.94	87.19	0.844	87.82	10.68
Proposed Method (#14)	**86.53**	90.23	**0.850**	**90.54**	**10.00**

TABLE VI

QUANTITATIVE COMPARISON BETWEEN THE SELECTED DEEP-LEARNING METHODS AND PROPOSED APPROACH WITH THE SVM CLASSIFIER FOR DATA-4

Methods	OA	AA	Ka	F1-score	SDUA
CNN [23]	71.60	71.79	0.685	65.27	23.62
SSRN-Net [25]	67.49	71.32	0.646	69.44	28.62
DPY-ResNet [26]	67.72	71.22	0.644	62.51	25.93
ASSM-Net [31]	75.90	71.55	0.730	64.94	16.20
SSR-Net [39]	53.71	49.48	0.471	30.32	25.76
Res-Net [34]	62.77	**73.71**	0.591	59.63	24.95
A2S2K-Net [33]	75.46	69.26	0.725	62.38	19.04
DFSL-Net [35]	74.64	73.41	0.719	69.96	20.00
MDL-Net [37]	75.34	68.75	0.725	68.22	25.14
S3-Net [36]	74.75	72.75	0.721	69.25	18.82
Proposed Method (#13)	**77.20**	70.57	**0.746**	**70.48**	**16.06**

sample enhancement approaches, including Richard's method [16] and Kang's method [21]. Moreover, the proposed approach achieved superior performance without the selection of any parameter. The improvement achieved by our proposed approach is indeed in terms of the four datasets and five evaluation indicators. The corresponding visual performance with less noise in Figs. 4–7 effectively supported the quantitative comparative conclusion, as indicated by the rectangles in the figures.

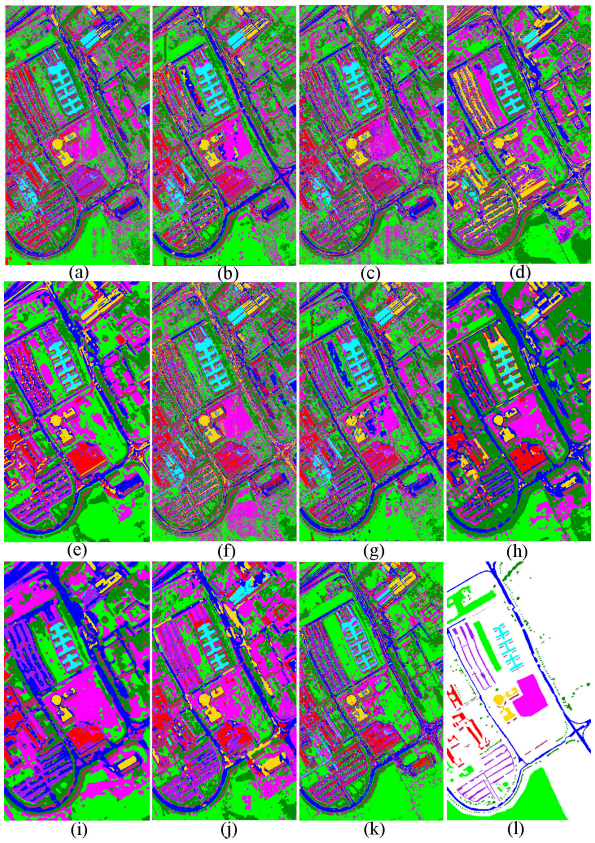


Fig. 8. Classification maps acquired with different methods for Data-1. (a) CNN [23], (b) SSRN-Net [25], (c) DPY-ResNet [26], (d) ASSM-Net [31], (e) SSR-Net [39], (f) Res-Net [34], (g) A2S2K-Net [33], (h) DFSL-Net [35], (i) MDL4OW-Net [37], (j) S3-Net [36], (k) our proposed approach, and (l) the ground-truth map.

2) The comparative results between deep-learning networks and our proposed approach are presented in Tables III–VI. The quantitative observations indicated that our proposed approach has an advantage in improving classification accuracies. Compared with the other methods in terms of Data 1, our proposed approach achieved the best OA (78.34%), Ka (0.719), F1 score (78.55%), and SDUA (12.05). Furthermore, the AA of our proposed approach is 76.83%, which is good as the best AA (77.64%) of A2S2K-Net [33]. The visual performance comparisons in Figs. 8–11 supported the quantitative observation conclusion, as shown in the highlighted parts of these figures. We can find that the visual performance obtained by our proposed approach has less noise than that of the others.

D. Discussion

The above experimental results demonstrated that the proposed approach is superior to the approach that used the initial samples directly and the widely used sample augmentation methods, including Richard's [16] and Kang's [21] methods. Moreover, the performance of the proposed approach has some advantages than that of DL-based methods, including SSRN-Net [25], CNN [23], DPY-ResNet [26], SSR-Net [39], ASSM-Net [31], Res-Net [34], A2S2K-Net [33], DFSL-Net [35], MDL4OW [37], and S3-Net [36]. To promote the

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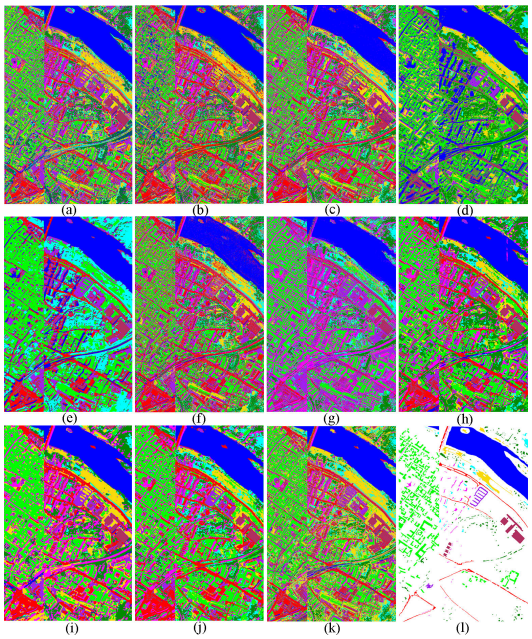


Fig. 9. Classification maps acquired with different methods for Data-2. (a) CNN [23], (b) SSRN-Net [25], (c) DPY-ResNet [26], (d) ASSM-Net [31], (e) SSR-Net [39], (f) Res-Net [34], (g) A2S2K-Net [33], (h) DFSL-Net [35], (i) MDL4OW-Net [37], (j) S3-Net [36], (k) our proposed approach, and (l) the ground-truth map.

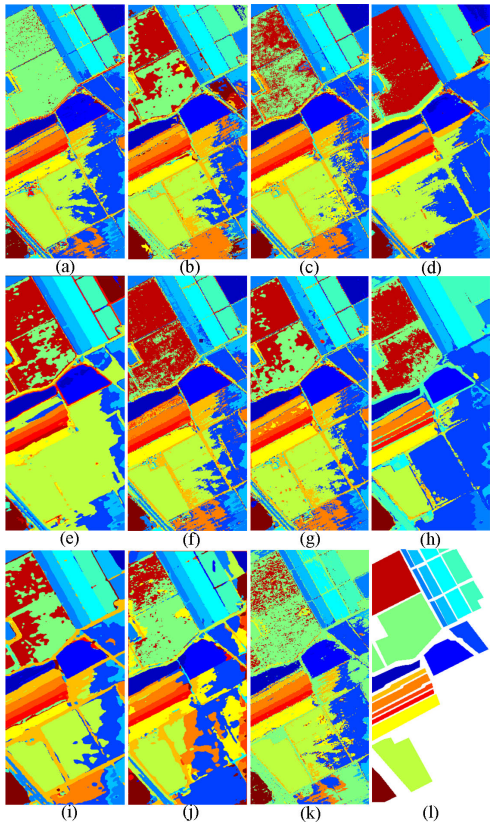


Fig. 10. Classification maps acquired with different methods for Data-3. (a) CNN [23], (b) SSRN-Net [25], (c) DPY-ResNet [26], (d) ASSM-Net [31], (e) SSR-Net [39], (f) Res-Net [34], (g) A2S2K-Net [33], (h) DFSL-Net [35], (i) MDL4OW-Net [37], (j) S3-Net [36], (k) our proposed approach, and (l) the ground-truth map.

widespread application of the proposed approach, we discussed its characteristics for the adaptive ability to different

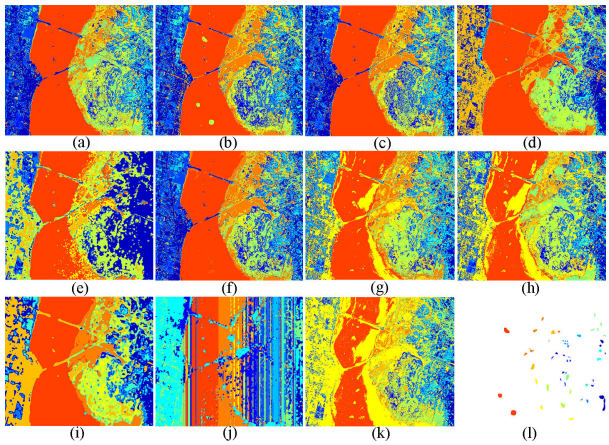


Fig. 11. Classification maps acquired with different methods for Data-4. (a) CNN [23], (b) SSRN-Net [25], (c) DPY-ResNet [26], (d) ASSM-Net [31], (e) SSR-Net [39], (f) Res-Net [34], (g) A2S2K-Net [33], (h) DFSL-Net [35], (i) MDL4OW-Net [37], (j) S3-Net [36], (k) our proposed approach, and (l) the ground-truth map.

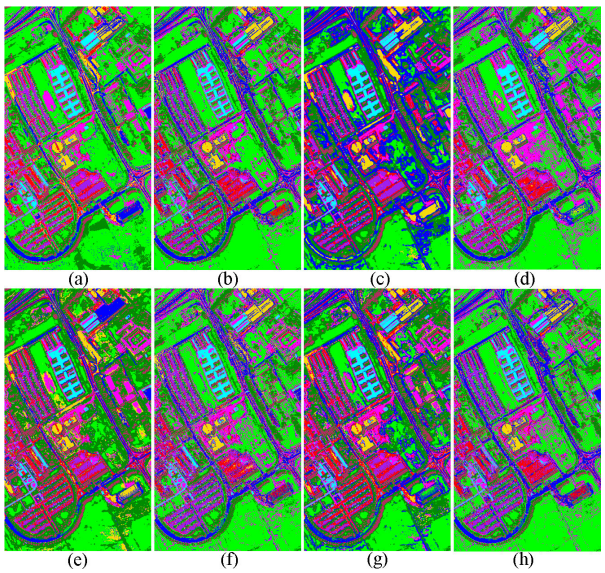


Fig. 12. Demonstration of visual comparisons between the classification map based on using initial samples and our proposed NLSA for different classifiers. (a) and (b) Obtained using the initial sample and our proposed approach with SVM, respectively. (c) and (d) For KNN. (e) and (f) For MLC. (g) and (h) For RF.

classifiers, the relationship between classification accuracies and iterations for different datasets, and the balancing ability for user accuracy.

First, the results based on the proposed approach, coupled with the widely used SVM, K-nearest neighbors (KNNs), maximum likelihood classifier (MLC), and random forest (RF) classifiers were presented in Table VII. The investigation and comparisons showed that the proposed approach was robust to the different classifiers and could improve the classification accuracies compared with using the initial samples directly. Moreover, the different iteration numbers of our proposed approach vary for different datasets and classifiers, but all achieved an enhanced classification performance. This phenomenon indicated that our proposed approach was robust and adaptive to different datasets and classifiers. Here, we selected

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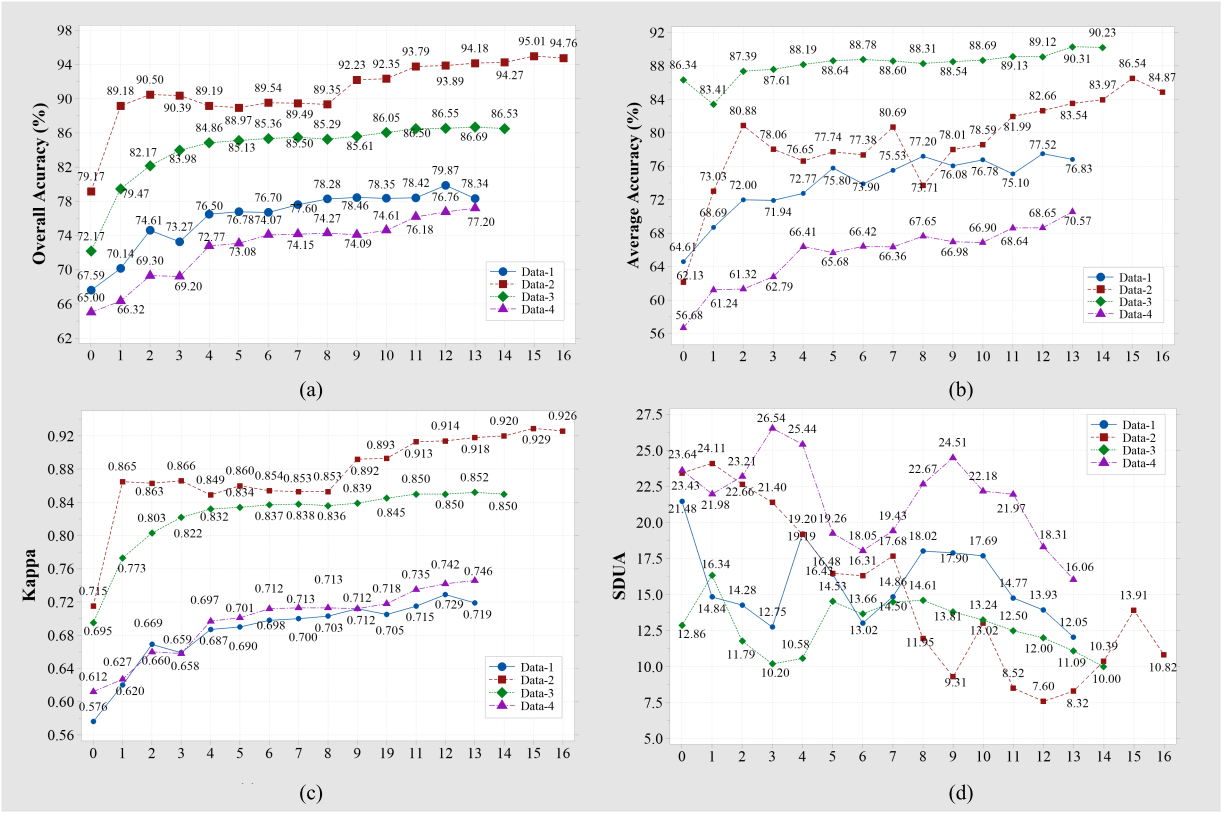


Fig. 13. Relationship between classification accuracies and iterations of the proposed approach coupled with SVM for Data-1. (a) Overall Accuracy, (b) Average Accuracy, (c) Kappa coefficient, (d) SDUA.

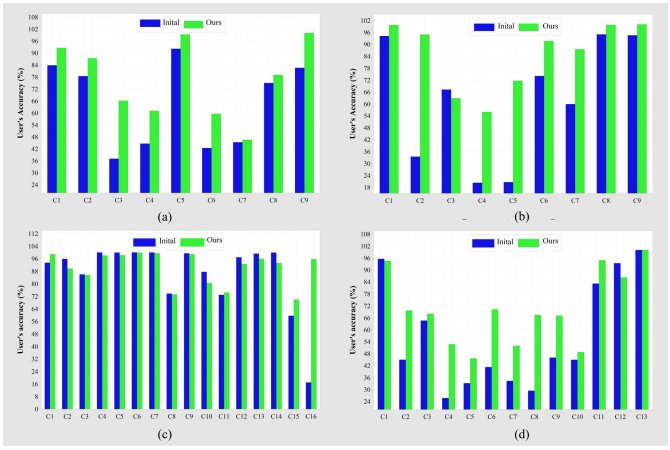


Fig. 14. Comparisons between results of using the initial training samples andthe enhanced samples with our proposed approach for the user accuracy of each classand dataset. (a) Data-1, (b) Data-2, (c) Data-3, (d) Data-4.

the visual performance on Data-1 for demonstration (see Fig. 12). The visual performance between the results based on the initial samples and our proposed approach for the four widely used classifiers indicated that our proposed approach can explore the high-quality potential samples without any parameter settings and the augmented samples are robust to the different classifiers.

Second, we investigate the relationship between classification accuracies and the number of iterations of the proposed approach for different datasets. As shown in Fig. 13: 1) the accuracies are first increased with the increment of iteration, then the accuracy tends to be at a stable level and 2) the total iterations of the proposed approach may be different for the same dataset and classifier. In addition, the change magnitude of SDUA is large and decreases with the increment of iteration because a sample change in one class can result in a large difference in a later classification for other classes.

Third, we also investigated the balancing ability of the proposed approach for the user's accuracy of each class. One demonstration based on SVM and Data 1 is shown in Fig. 14, which indicates that the proposed approach can balance the user's accuracy by adjusting automatically the samples of each class. The quantitative comparisons between the results based on the initial samples and our proposed approach in terms of SDUA supported the observed conclusion.

Fourth, the relationship between classification accuracies and the number of selection bands of HSRIs while using the few-shot learning neural networks (DFSL-Net [35], MDL4OW [37], S3-Net [36]) and our proposed approach. Fig. 15 for Data-1 and Fig. 16 for Data-3 indicate that the classification accuracies increase gradually when the number of selected bands is increased from 3 to 10, and all the approaches reach the best accuracies when the number of selected bands is 10 or 11. However, after that, the classification accuracies decrease with increasing selected bands. That is because the intraclass variance increases and the interclass variance decreases, resulting in a decrease in the separability among different targets when the selected bands are increased for the classification of HRSIs. In the same way, a similar observation conclusion can be concluded. Moreover, it is worth noting that the proposed approach achieves higher accuracies

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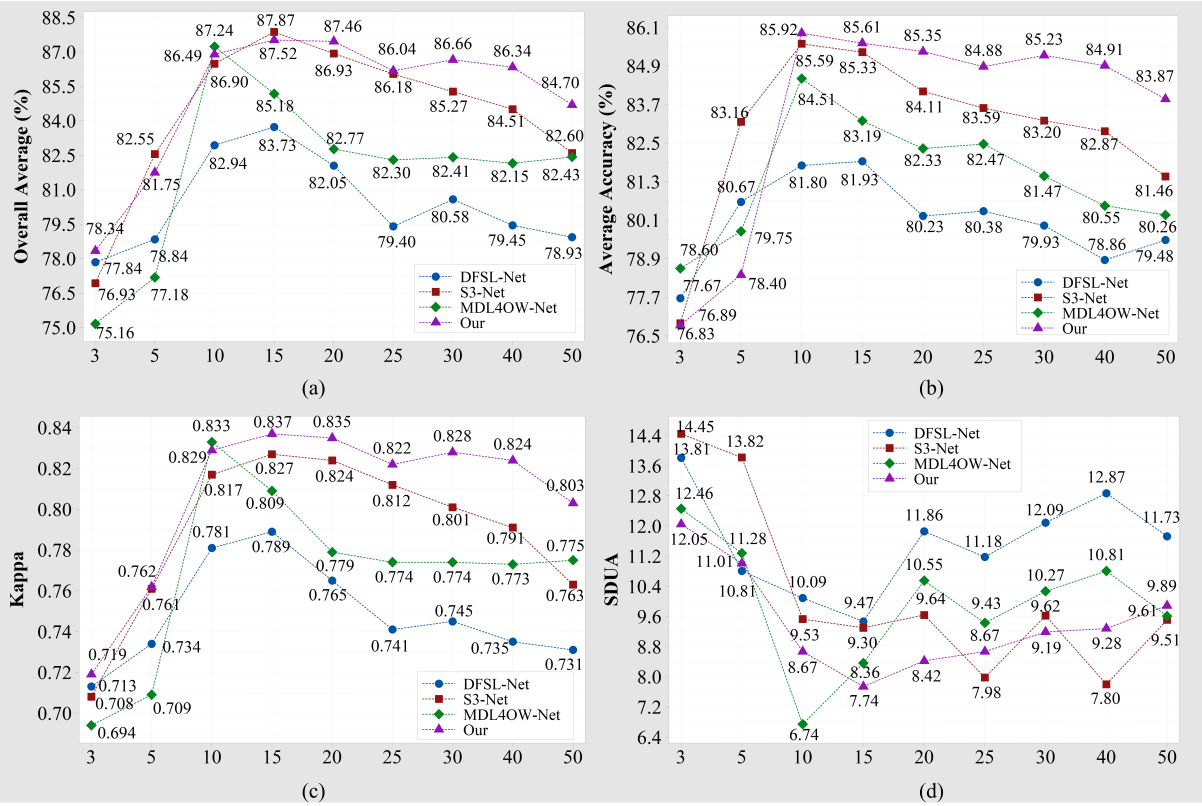


Fig. 15. Relationship between classification accuracies and the number of bands for Data-1and each method. (a) Overall Accuracy, (b) Average Accuracy, (c) Kappa coefficient, (d) SDUA.

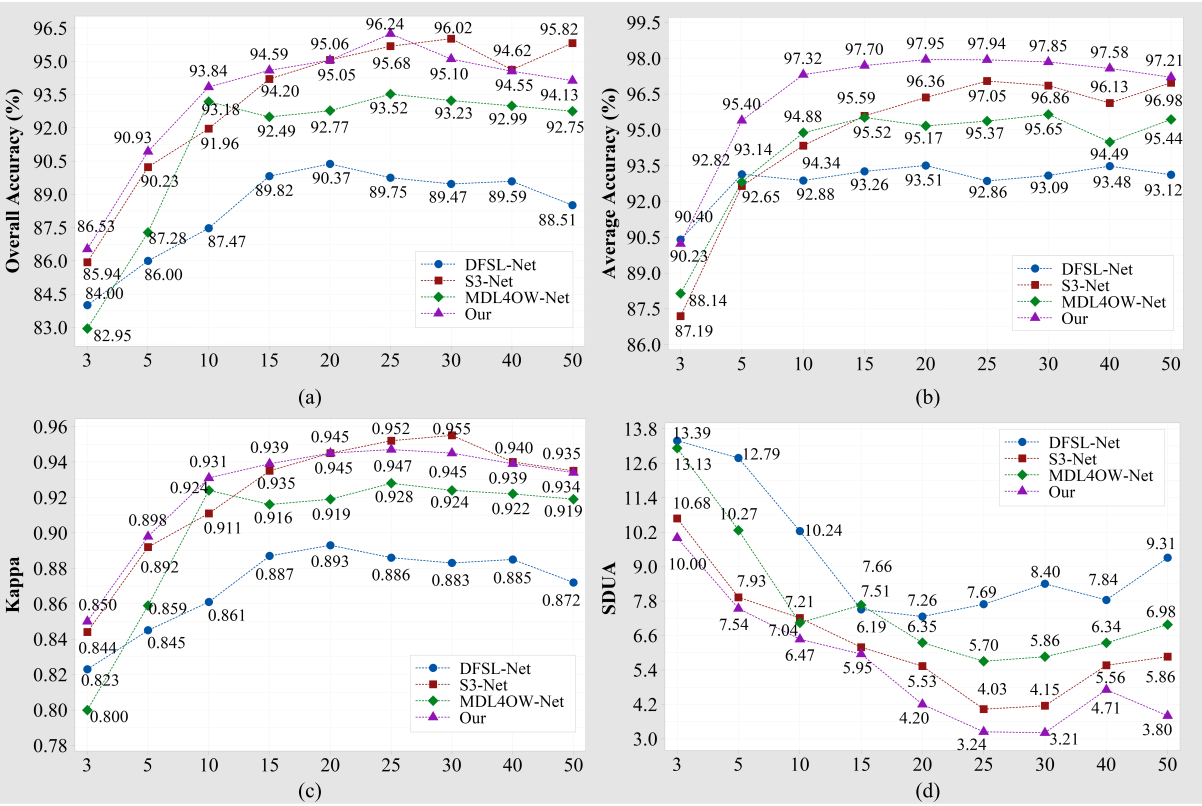


Fig. 16. Relationship between classification accuracies and the number of bands for Data-3 and each method. (a) Overall Accuracy, (b) Average Accuracy, (c) Kappa coefficient, (d) SDUA.

at the same selected bands when compared with DFSL-Net [35], MDL4OW [37], and s3-Net [36].

In addition, the computational complexity of the proposed approach and the selected deep-learning methods are evaluated

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TABLE VII

PERFORMANCE OF OUR METHOD IN COMPARING THE CLASSIFICATION ACCURACY OF THE INITIAL SAMPLE OF THE CLASSIFIER WITH THAT OF THE TERMINAL SAMPLE (# NUMBER DENOTES THE TOTAL ITERATIONS OF PROPOSED APPROACH, Ka $\in [-1, +1]$, OTHERS MEASURED IN PERCENT)

Dataset	Classifier	OA		AA		Ka		F1-score		SDUA	
		Initial	Proposed	Initial	Proposed	Initial	Proposed	Initial	Proposed	Initial	Proposed
Data-1	SVM (#13)	67.59	**78.34**	64.61	**76.83**	0.575	**0.719**	65.37	**78.55**	21.48	**12.05**
	KNN (#15)	64.18	**78.35**	60.72	**75.45**	0.544	**0.709**	58.82	**73.06**	24.00	**12.44**
	MLC (#14)	63.63	**78.65**	62.65	**76.76**	0.543	**0.725**	60.07	**78.72**	23.42	**10.77**
	RF (#13)	66.60	**80.51**	61.63	**78.22**	0.574	**0.736**	61.92	**76.13**	22.49	**14.25**
Data-2	SVM (#16)	79.17	**94.76**	62.13	**84.87**	0.715	**0.926**	64.00	**86.32**	23.43	**10.82**
	KNN(#15)	76.49	**93.09**	65.92	**81.67**	0.675	**0.903**	62.94	**82.96**	22.29	**10.77**
	MLC (#14)	77.78	**91.56**	63.69	**77.98**	0.694	**0.882**	63.53	**78.88**	19.98	**13.36**
	RF(#13)	81.03	**93.42**	66.39	**80.27**	0.733	**0.908**	66.01	**82.91**	19.69	**9.34**
Data-3	SVM (#14)	72.17	**86.53**	86.34	**90.23**	0.695	**0.850**	79.37	**90.54**	12.86	**10.00**
	KNN (#13)	74.51	**86.83**	78.31	**90.80**	0.718	**0.853**	76.30	**90.02**	21.87	**9.98**
	MLC (#13)	77.18	**88.47**	79.95	**91.79**	0.747	**0.871**	79.97	**90.74**	18.86	**9.95**
	RF (#12)	80.93	**85.93**	83.88	**89.83**	0.789	**0.843**	85.30	**90.30**	11.74	**11.67**
Data-4	SVM (#13)	65.00	**77.20**	56.68	**70.50**	0.612	**0.746**	56.05	**70.48**	23.64	**16.06**
	KNN (#14)	63.02	**76.49**	54.89	**68.03**	0.590	**0.739**	53.95	**68.55**	24.21	**18.53**
	MLC (#13)	65.92	**77.61**	58.03	**70.52**	0.621	**0.751**	57.96	**70.64**	22.69	**16.08**
	RF (#14)	61.97	**76.59**	52.62	**68.90**	0.578	**0.739**	52.35	**68.06**	24.01	**20.86**

TABLE VIII

COMPARISON BETWEEN OUR PROPOSED METHOD COUPLED WITH THE SVM CLASSIFIER AND THE SELECTED DEEP-LEARNING METHODS IN TERMS OF TIME-COST. (# NUMBER DENOTES THE TOTAL ITERATIONS OF PROPOSED APPROACH, THE UNIT IS MINUTE)

Methods	Data-1	Data-2	Data-3	Data-4
CNN [23]	37.5	93.2	32.5	42.6
SSRN-Net [25]	53.1	97.6	46.0	58.3
DPY-ResNet [26]	48.9	72.4	44.2	57.3
ASSM-Net [31]	155.8	328.1	110.8	165.8
SSR-Net [39]	49.9	66.6	41.6	52.3
Res-Net [34]	56.3	79.2	47.7	53.2
A2S2K-Net [33]	37.5	93.2	32.5	42.6
DFSL-Net [35]	123.6	227.5	114.55	189.2
MDL4OW-Net [37]	82.7	138.3	77.6	126.3
S3-Net [36]	69.6	127.3	66.2	117.2
Proposed Method	(#13)261.2	(#16)481.4	(#14)371.6	(#13)289.9

for each dataset. As shown in Table VIII, all the approaches were conducted using Intel[1] Core i7-7700 CPU with 3.60 GHz and 1080Ti GPU. The observations indicate that the proposed approach requires more time to achieve a better classification performance, because the proposed approach was an iterative algorithm that needs more time to explore potential samples around each initial sample and balance the user's accuracy of each class, and the proposed method is less time-efficient. This is a limitation of our proposed approach for practical application.

IV. CONCLUSION

In this article, a novel land-cover classification approach with sample augmentation is proposed to improve classification accuracy and balance the deviation of the user's accuracy for HRSIs. To achieve these objectives, the proposed method assumes that the spectral feature in a land-cover patch is typically heterogeneous in HRSIs and attempts to enrich and adjust the training samples for each class during the supervised classification process. The proposed approach is implemented with a very small initial training sample and without parameter hard-tuning, but it achieves higher accuracy compared with some cognate methods. In addition, the proposed approach exhibits an advantage in improving the user's accuracy of nearly all the classes and balancing deviation among each user's accuracy. Thus, the proposed approach may be preferred by many practitioners.

The effectiveness and adaptability of the proposed approach are validated on four real HRSIs acquired from different sensors, platforms, and resolutions. The study areas are typical urban and rural scenes with various land-cover types, including roads, trees, meadows, and buildings. Quantitative evaluation and visual comparisons based on the images indicate that the proposed approach can achieve higher accuracy and balance the user's accuracy for different classes.

In summary, the proposed approach is a promising algorithm for HRSI classification. From the methodological perspective, however, the proposed approach still requires additional research. In our future study, we plan to apply it to large areas and other types of HRSIs to further test the robustness and adaptability of the proposed approach. In addition, convolutional multiscale features will be adopted instead of the single spectral feature used in the proposed approach to improve its robustness and effectiveness.

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ACKNOWLEDGMENT

The authors appreciate the editor-in-chief, associate editor, and reviewers for their insightful comments and suggestions.

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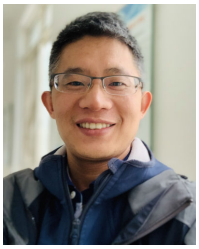
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Zhiyong Lv (Senior Member, IEEE) received the M.S. and Ph.D. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2008 and 2014, respectively.

He was an Engineer in surveying and worked at The First Institute of Photogrammetry and Remote Sensing, Xi'an, China, from 2008 to 2011. He is currently with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an. His research interests include multihyperspectral and high-resolution remotely sensed image processing, spatial feature extraction, neural networks, pattern recognition, deep learning, and remote-sensing applications.



Pengfei Zhang is currently pursuing the master's degree in computer science with the Xi'an University of Technology, Xi'an, China.

His research interests include spatial–spectral feature extraction, pattern recognition, ground target detection, and land-cover/land-use change detection, through remote-sensing images with a high or very high spatial resolution (including satellite imagery and aerial images).



Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

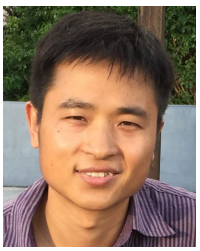
From 2011 to 2012, he was with the Department of Applied Mathematics, University of Maryland at College Park, College Park, MD, USA, as a Visiting Scholar with the Famous Professor John Benedetto to study the dimensionality reduction of hyperspectral images. From 2014 to 2016, he studied with the State Key Laboratory for Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, as a Post-Doctoral Fellow to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is currently a Full Professor with Ningbo University, Ningbo, China. He has authored more than 70 journal articles. His research interests include hyperspectral image processing with manifold learning, anomaly detection, and target recognition of remote-sensing imagery using compressive sensing.



Jón Atli Benediktsson (Fellow, IEEE) received the Cand.Sci. degree in electrical engineering from the University of Iceland, Reykjavik, Iceland, in 1984, and the M.S.E.E. and Ph.D. degrees from Purdue University, West Lafayette, IN, USA, in 1987 and 1990, respectively.

He is currently a Pro-Rector for Academic Affairs and a Professor of electrical and computer engineering with the University of Iceland. His research interests include remote sensing, biomedical analysis of signals, pattern recognition, image processing, and signal processing. He has authored extensively in these fields.

Dr. Benediktsson is a fellow of Society of Photo-Optical Instrumentation Engineers (SPIE). He received the Stevan J. Kristof Award from Purdue University in 1991 as an outstanding graduate student in remote sensing. In 1997, he was a recipient of the Icelandic Research Council's Outstanding Young Researcher Award. In 2000, he was granted the IEEE Third Millennium Medal. In 2004, he was a corecipient of the University of Iceland's Technology Innovation Award. In 2006, he received the Yearly Research Award from the Engineering Research Institute of the University of Iceland. In 2007, he received the Outstanding Service Award from the IEEE Geoscience and Remote Sensing Society. He was a corecipient of the 2012 IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) Paper Award. In 2013, he was a corecipient of the IEEE GRSS Highest Impact Paper Award. In 2013, he also received the IEEE/VFI Electrical Engineer of the Year Award. He is a member of the Association of Chartered Engineers in Iceland (VFI), Societas Scinetiarum Islandica, and Tau Beta Pi. He was the 2011–2012 President of the IEEE Geoscience and Remote Sensing Society (GRSS) and has been on the GRSS AdCom since 2000. He was an Editor of the IEEE TGRS from 2003 to 2008 and has been serving as an Associate Editor of TGRS since 1999, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS since 2003, and IEEE ACCESS since 2013. He is on the International Editorial Board of the *International Journal of Image and Data Fusion*. He was the Chair of the Steering Committee of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (J-STARS) from 2007 to 2010. He is a Co-Founder of the biomedical startup company Oxymap (www.oxymap.com).



Tao Lei (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from Northwestern Polytechnical University, Xi'an, China, in 2011.

From 2012 to 2014, he was a Post-Doctoral Research Fellow with the School of Electronics and Information, Northwestern Polytechnical University. From 2015 to 2016, he was a Visiting Scholar with the Quantum Computation and Intelligent Systems Group, University of Technology Sydney, Sydney, NSW, Australia. From July 2017 to October 2017, he was a Research Fellow with the College of Electronic and Computer Engineering, Brunel University London, Uxbridge, U.K. He is currently a Professor with the School of Electronic Information and Artificial Intelligence, Shaanxi University of Science and Technology, Xi'an. He has authored or coauthored more than 80 research articles, including IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON FUZZY SYSTEMS (TFS), and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS). His research interests include image processing, pattern recognition, and machine learning.

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